

DNA from the growing embryo. How does it get there? The mother's blood reaches the embryo through the placenta. When cells produced by the embryo grow old, die, and break open within the placenta, the released DNA enters the mother's circulation. Although the blood also contains pieces of DNA from the mother, about 10–15% of the DNA circulating in the blood is from the fetus. Both the polymerase chain reaction (PCR) and high-throughput sequencing can convert the bits of fetal DNA into useful information.

Unfortunately, almost all detectable disorders remain untreatable in the uterus, and many cannot be corrected even after birth. Genetic testing may leave parents faced with difficult decisions about whether to terminate a pregnancy or to raise a child who may have profound developmental abnormalities and a short life expectancy. These are complex issues that demand careful, informed thought and competent genetic counseling.

Parents will be receiving even more genetic information and confronting further questions in the near future. Indeed, in 2012 the first infant was born whose entire genome had been sequenced before the baby was born.

Nevertheless, completing a genome sequence does not ensure complete information. Consider, for example, Klinefelter syndrome, in which males have an extra X chromosome. This disorder is quite common, affecting one in 1,000 men, and can cause reduced testosterone, a feminized appearance, and infertility. However, while some men with an extra X chromosome have a debilitating disorder, others have symptoms so mild that they are unaware of the condition. For other disorders, such as diabetes, heart disease, or cancer, a genome sequence may only indicate the degree of risk. How parents will use this and other information in having and raising children is a question with no clear answers.

CONCEPT CHECK 46.5

1. Why does testing for hCG (human chorionic gonadotropin) work as a pregnancy test early in pregnancy but not late in pregnancy? What is the function of hCG in pregnancy?
2. In what ways are tubal ligation and vasectomy similar?
3. **WHAT IF?** If a sperm nucleus is injected into an oocyte, what steps of gametogenesis and conception are bypassed?

For suggested answers, see Appendix A.

46 Chapter Review

SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zk9t9t.

CONCEPT 46.1

Both asexual and sexual reproduction occur in the animal kingdom (pp. 1020–1022)

- **Sexual reproduction** requires the fusion of male and female gametes, forming a diploid **zygote**. Examples of **asexual reproduction**—the production of offspring without gamete fusion—include budding, **fission**, and fragmentation with regeneration. Variations on the mode of reproduction are achieved through **parthenogenesis**, **hermaphroditism**, and sex reversal. Hormones and environmental cues control reproductive cycles.

? Would a pair of haploid offspring produced by parthenogenesis be genetically identical? Explain.

CONCEPT 46.2

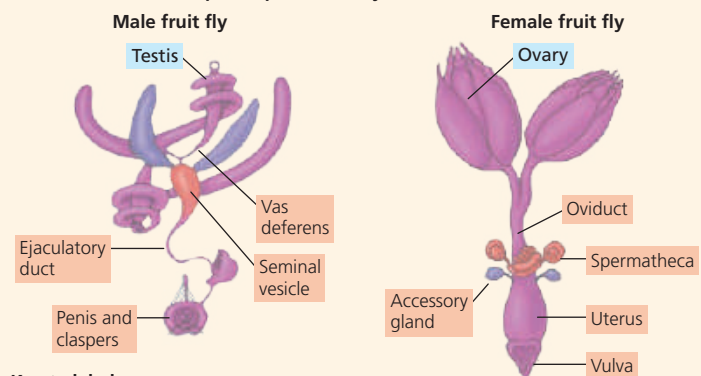
Fertilization depends on mechanisms that bring together sperm and eggs of the same species (pp. 1022–1025)

- **Fertilization** occurs externally, when sperm and eggs are both released outside the body, or internally, when sperm deposited by the male fertilize an egg in the female reproductive system. In either case, fertilization requires coordinated timing, which may

be mediated by environmental cues, pheromones, or courtship behavior. Internal fertilization is often associated with relatively fewer offspring and greater protection of offspring by the parents.

- Systems for gamete production and delivery range from undifferentiated cells in the body cavity to complex systems that include **gonads**, which produce gametes, and accessory tubes and glands that protect or transport gametes and embryos. Although sexual reproduction involves a partnership, it also provides an opportunity for competition between individuals and between gametes.

Complex reproductive systems in fruit flies



Key to labels:

Gamete production

Gamete protection and transport

? Identify which of the following, if any, are unique to mammals: a female uterus, a male vas deferens, extended internal development, and parental care of newborns.